
DETECTING HIDDEN BIASES IN CLINICAL AI: A FRAMEWORK FOR FAIR STROKE OUTCOME PREDICTION USING CLINICAL DATA

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LAY SUMMARY



FAIRNESS

The diagram features a central blue rectangular box with the word "FAIRNESS" in white, serif, all-caps font. Surrounding this box are four illustrations: a doctor in a white coat and stethoscope at the top, a patient lying down on the left, and an AI robot head with a yellow "AI" chip on its forehead at the bottom. Curved black arrows connect the doctor to the patient, the patient to the AI, and the AI back to the doctor, forming a clockwise cycle. A green rectangular box is positioned to the right of the AI robot head, with a curved black arrow pointing from it towards the top right corner of the diagram.

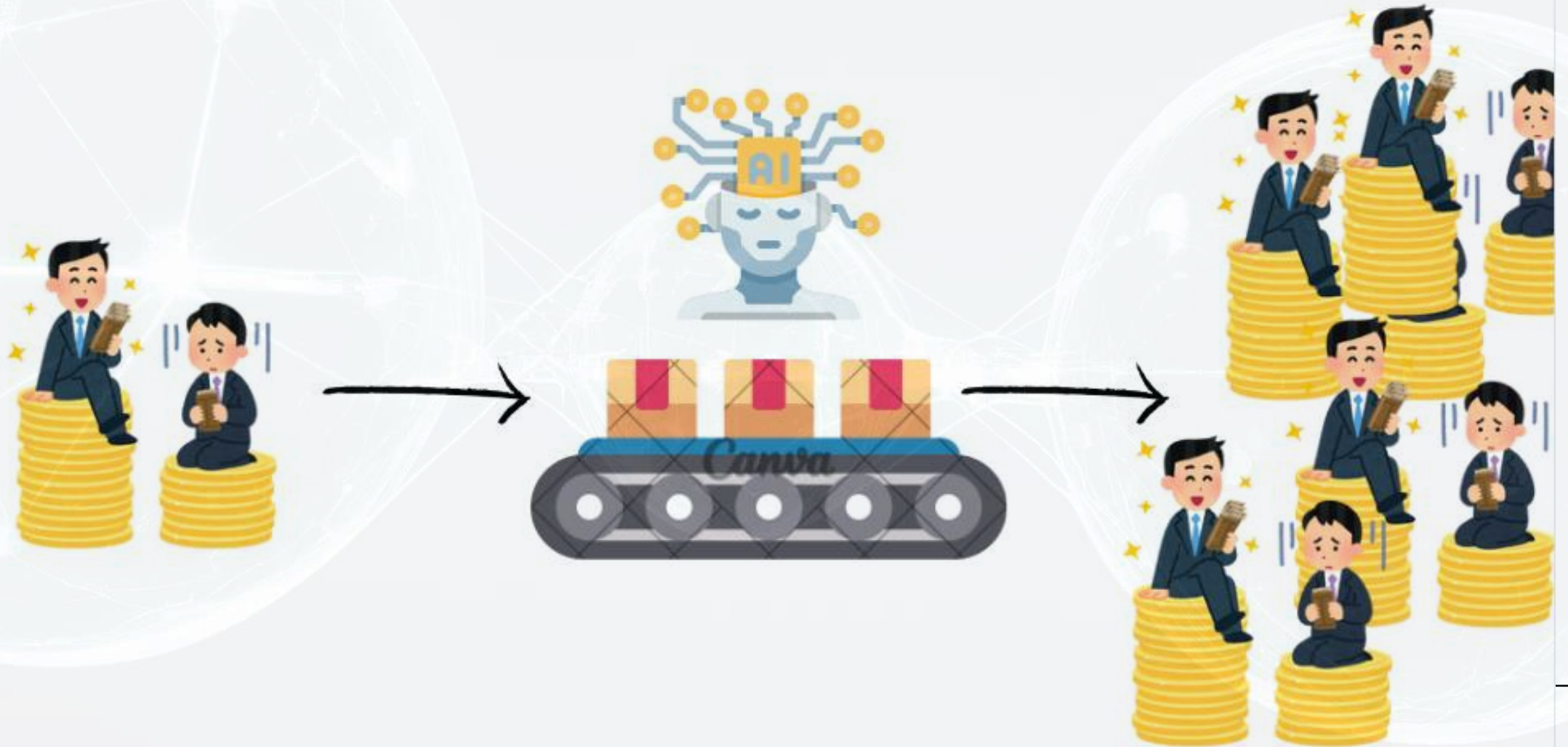
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BIAS:

In the context of **healthcare AI**, bias can be defined as any **systematic and/or unfair difference** in how predictions are generated for different patient populations that could lead to **disparate care delivery**¹

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PROBLEM



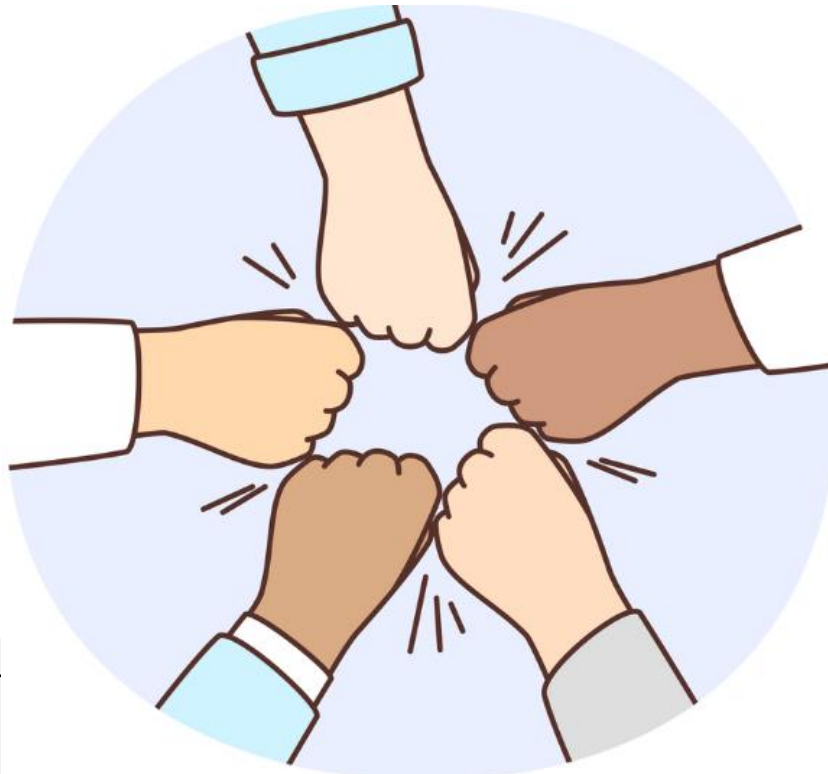
PREVIOUS WORK

FRAMEWORKS



LEGISLATION

50% of studies
developed models
with a high risk
of bias²



83% of
neuroimaging
studies developed
models with a
high risk of bias³



LAY SUMMARY

① BIAS **IMPACT**

How can the
IMPACT of
BIAS be
quantified?

How much does
BIAS affect the
performance of
a ML model

② BIAS **IMPROVEMENT**

Framework for
addressing **BIAS**
throughout a
model's lifecycle



CLINICAL IMPACT OF BIAS

- **Impact of Bias in Healthcare Models:**
 - **Healthcare Inequalities:** Biases can exacerbate existing disparities, particularly in stroke care, where **rural and low-income** communities face higher death rates.
 - **MIMIC-III & MIMIC-IV:** Studies show that stroke prediction models perform differently across demographic groups, especially for **rural patients** who lack access to diagnostic tools and treatments (Obermeyer et al., 2019).
 - **Consequences of Bias:**
 - **Delayed Treatment:** Biased models could delay urgent treatments, like **thrombectomy**, where timely intervention is critical (Saver, 2006).
 - **Erosion of Trust:** Lack of fairness in AI tools can reduce **trust** in healthcare, especially in high-stakes areas like stroke care.
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EVOLVING TREATMENT PRACTICES

- **The Changing Landscape of Stroke Treatment:**
 - **MIMIC Data Evolution:** MIMIC-III (2001-2012) and MIMIC-IV (2008-2019) reflect shifts in treatment practices, such as **thrombectomy**.
 - **Outdated Models:** Algorithms trained on old data may miss newer treatment opportunities, leading to **worse outcomes** and **higher healthcare costs**.





SOCIETAL & REGULATORY IMPLICATIONS

- **Regulatory and Societal Implications:**

- **FDA Fairness Checks:** Regulatory bodies now require **fairness checks** to ensure that AI models do not harm underrepresented groups.
- **Bias Detection Throughout Model Lifecycle:** Bias detection should occur **continuously** (data selection → model development → deployment) to prevent harmful outcomes.



PROPOSED METHOD & MACHINE LEARNING SOLUTION

Step I: Studying Bias and Its Types.

Step II: Choosing a model

Step III: Replicate the Study

**Step IV: Detect Biases and Generate a
Report**

STUDYING BIAS AND ITS TYPES

Step I: Studying Bias and Its Types

- **Historical Bias:** Labels reflect past inequities in medical diagnosis/treatment.
- **Measurement Bias:** Features act as proxies for protected attributes model and ZIP code representing black neighborhoods.
- **Representation Bias:** Key patient groups underrepresented in training data.

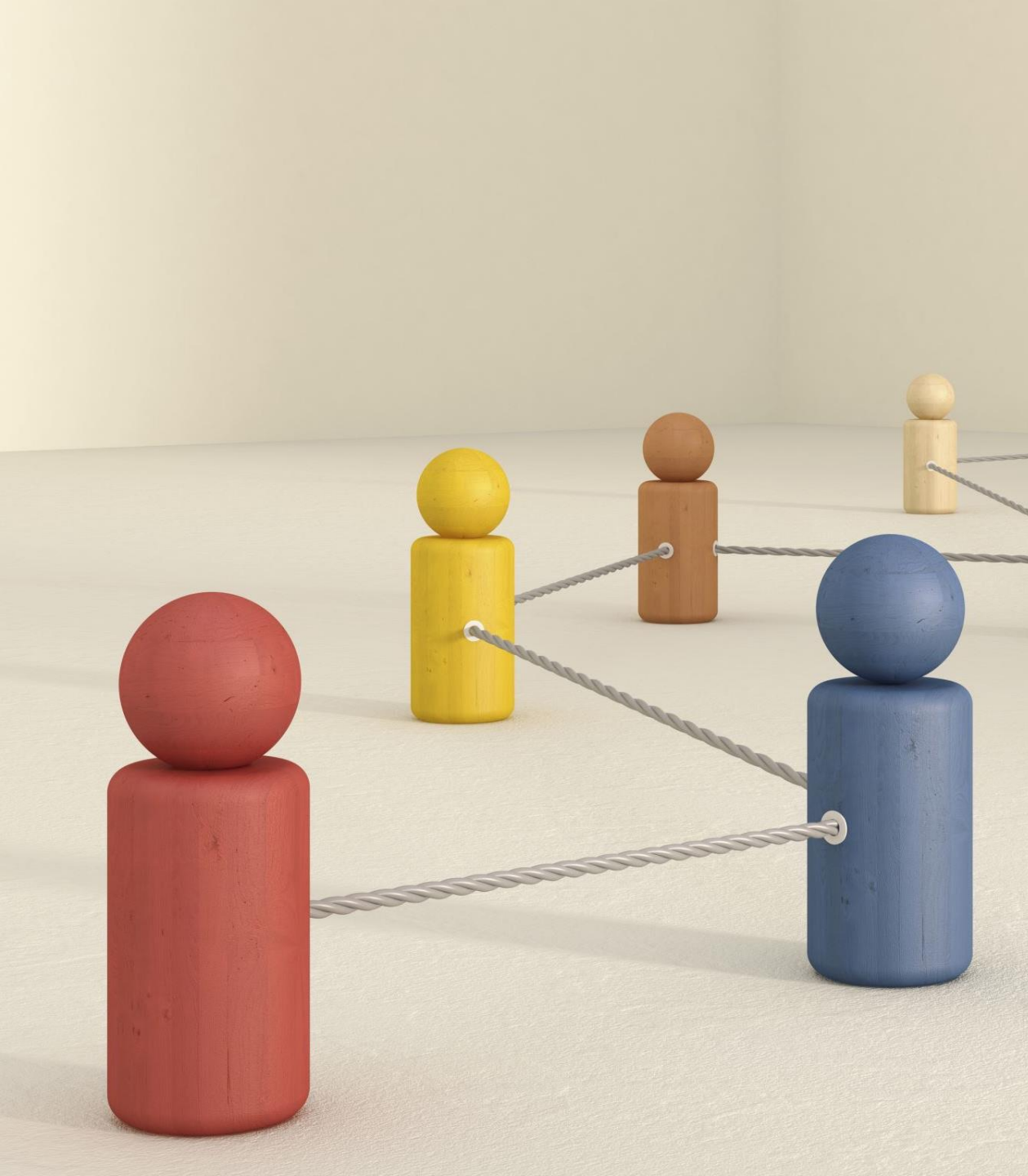
Algorithmic Bias: Model architecture harms specific groups.

Evaluation Bias: Metrics hide disparities across patient subgroups.

Interaction Bias: Harmful combinations of features disadvantage certain groups.

Aggregation Bias: Model ignores subgroup differences in medical thresholds.





CHOOSING A MODEL

Step II: Choosing a model

- The model should solve a critical classification problem in healthcare
 - The model should be a result of an authentic and reputable study
 - The training data of the model should have sufficient variables, such as sociodemographic indicators, to enable identification of subgroups; and records to enable testing for temporal variations
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REPLICATE STUDY & DETECT BIAS

Step III: Replicate



Replicate the Study

Follow the replication guidance provided in the paper of the selected study

Step IV: Detect

Detect Biases and Generate a Report

Test for the relevant biases in the replicated model and present a report highlighting subgroups in the data set that are affected by the bias, and the degree of their impact.

MACHINE LEARNING PREDICTION MODELS FOR POSTOPERATIVE STROKE IN ELDERLY PATIENTS

- Postoperative stroke in elderly patients is a significant concern due to aging populations and high stroke prevalence.
- Develop a reliable prediction model to identify high-risk patients early, enabling timely intervention and improved outcomes.

Medical Information Mart for Intensive Care

- **Longitudinal:** 2008–2022 data enables bias checks over time.
- **Rich Variables:** Demographics, labs, and comorbidities support subgroup analysis.





Long process to get access to MIMIC data sets on PhysioNet platform.

Requires filling out a form.

A referral from a supervisor under whom you're working.

Requires completion of Collaborative Institutional Training Initiative (CITI Program).



Massive Size of the Data Set

Processing the data is hard because it uses high computation power.



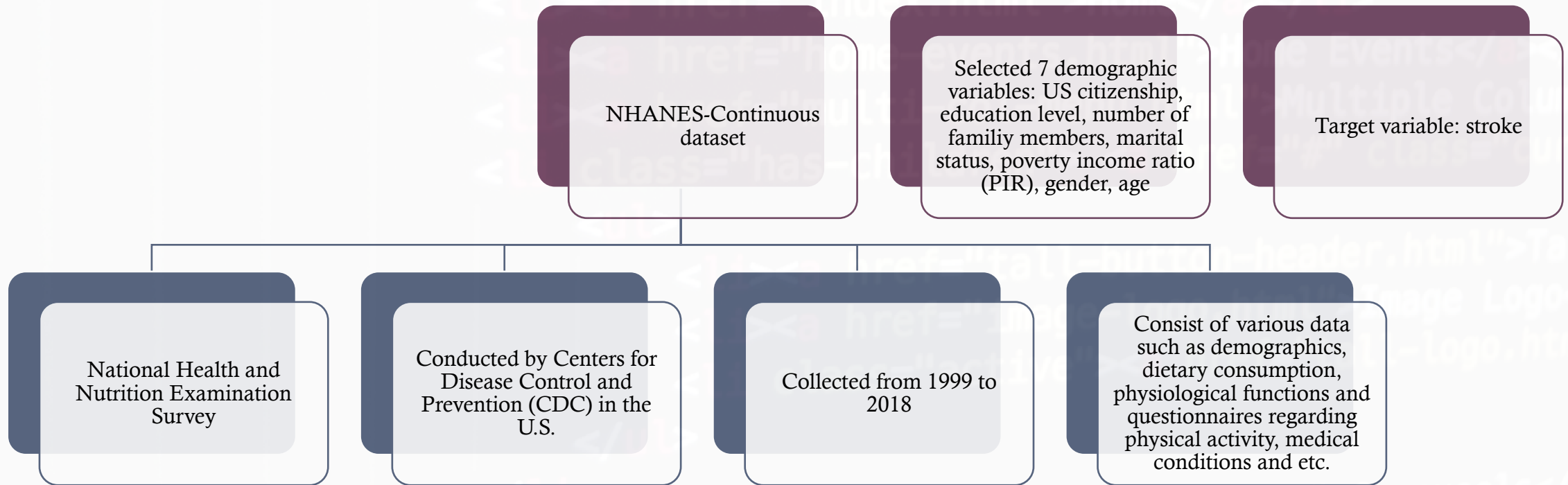
Absence of Explicit Guidance on Study Replication

There is no GitHub link or code provided to generate the training and testing data sets to replicate the study.

Therefore, a simpler dataset was required to test our methodology for bias detection.

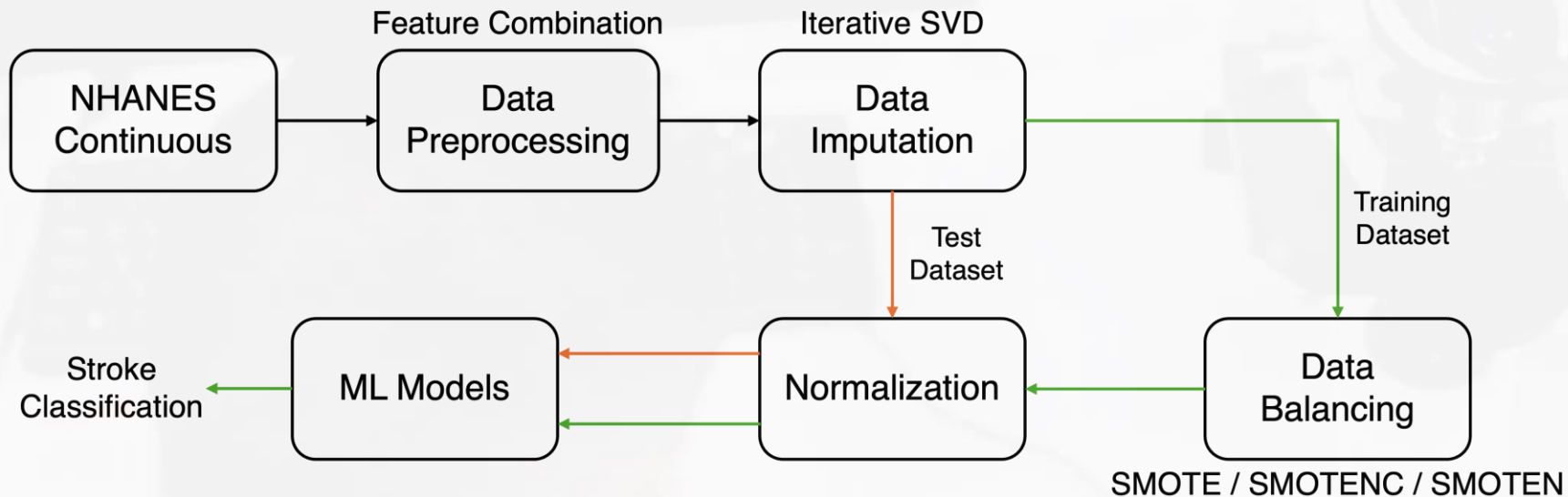
ANALYSES OF THE MIMIC DATABASE

PROOF-OF-CONCEPT: DATA

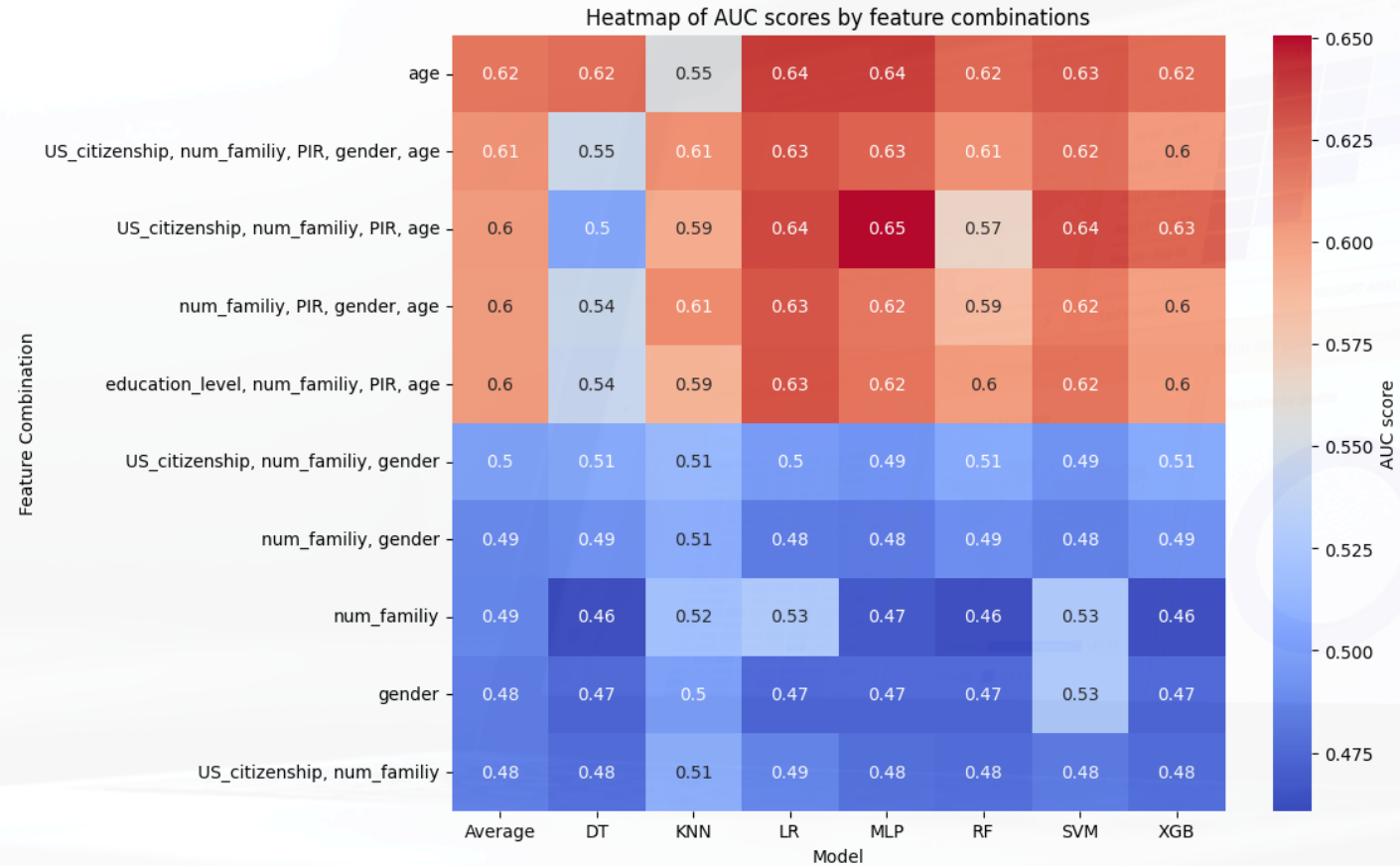


PROOF-OF-CONCEPT: WORKFLOW

- Target validation ML application: (Zhang et al., 2022)



PROOF-OF-CONCEPT: RESULTS



INTERDISCIPLINARY FRAMEWORK



- **Broad & Inclusive:**
 - **Goal:** Detect bias at all stages of healthcare AI development (from planning to post-deployment).
 1. **Stakeholder Involvement**
 2. **Interdisciplinary Collaboration**
 3. **Stakeholder Engagement**

Build a socially responsible and inclusive AI model that is not only technically sound but also addresses real-world health disparities.

INTERDISCIPLINARY COLLABORATION



Medical Historians



Data Scientists



Policy Makers

Planning

Development

Validation &
Evaluation

Post-Deployment



Social Scientist and Sociologist

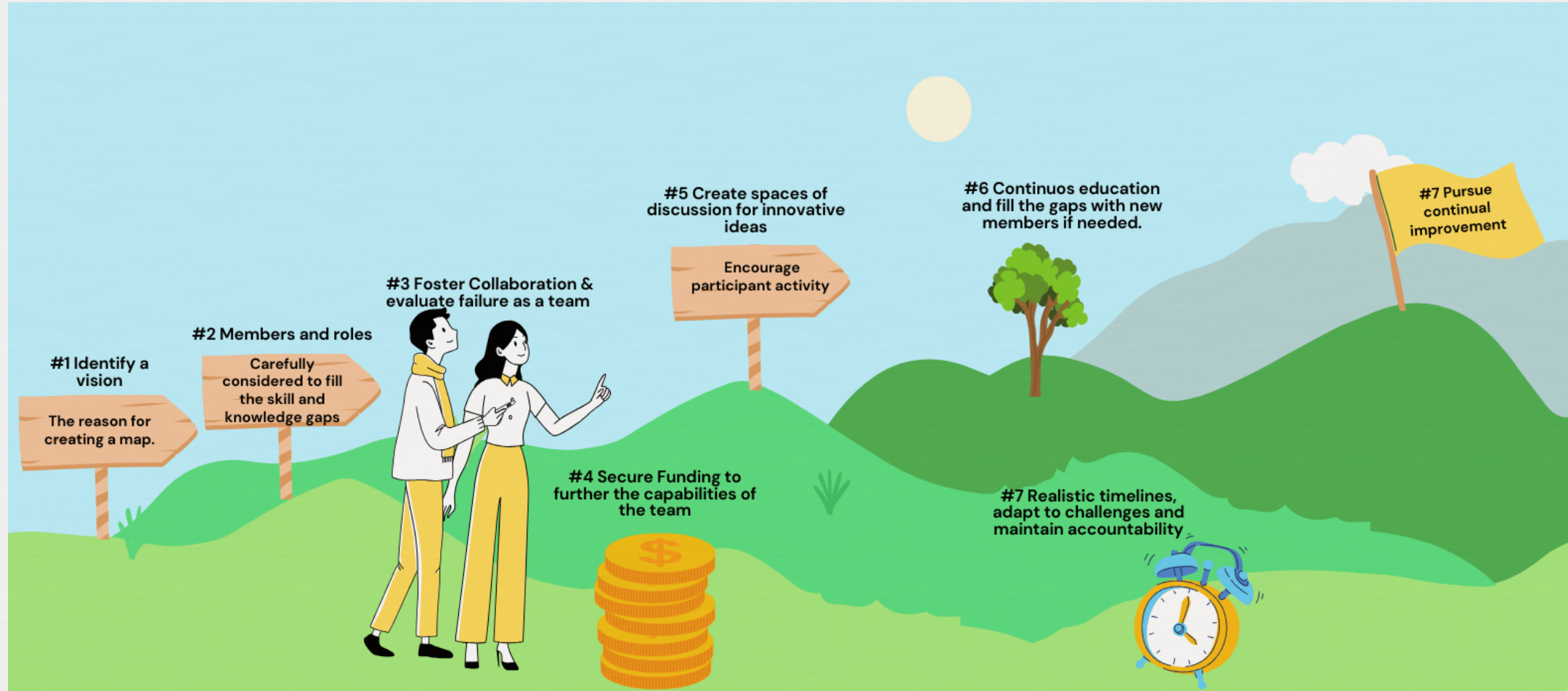


Underrepresented patients



Clinicians & Hospital workers

STAKEHOLDER ENGAGEMENT STRATEGIES



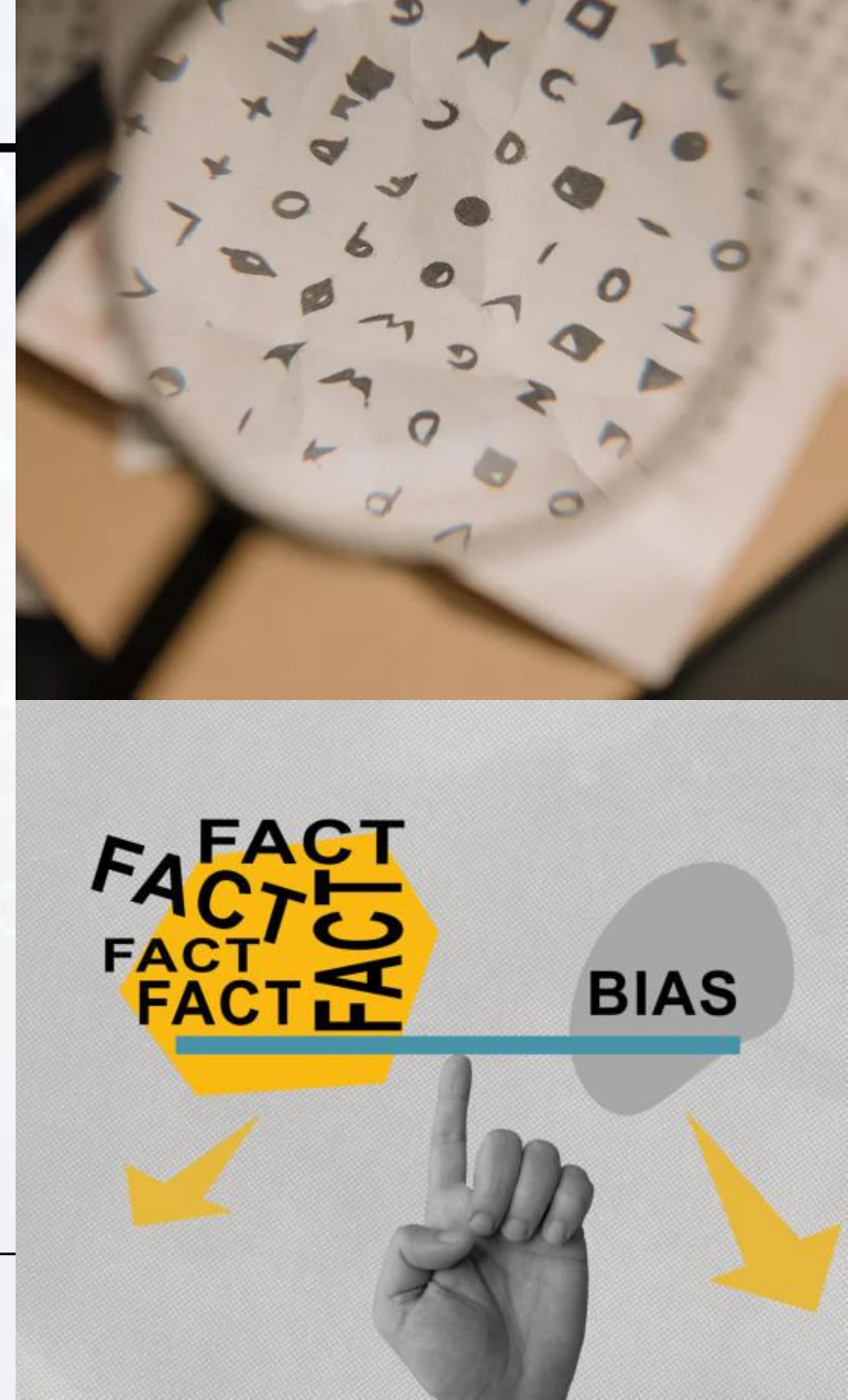
CURRENT LIMITATIONS & UNIQUE APPROACH

- **Current Bias Detection Limitations:**

- Existing tools (e.g., IBM Fairness 360) focus on post-deployment fixes.
- Biases often develop over time or due to evolving data (e.g., treatment advances or rural-urban disparities).
- Studies overlook changes like stroke treatment (Saver, 2006) and healthcare access differences (Obermeyer et al., 2019).

- **Unique Approach:**

- Bias detection throughout the *entire model lifecycle*: data selection, training, deployment.
- Examines biases arising from data shifts (e.g., rural vs. urban healthcare access).





INNOVATIVE METHODS & REDEFINING SUCCESS

- **Innovative Methods:**
 - **Synthetic data** simulates real-world scenarios (e.g., delayed access to neuroimaging in rural hospitals).
 - *Collaborative framework* with neurologists defining fairness criteria and data scientists implementing them as measurable metrics.
 - **Flexible Framework:**
 - Adapts to diverse settings (academic research, community hospitals).
 - Focus on relevant biases for specific settings: e.g., health coverage status in resource-limited areas, or treatment changes between MIMIC-III and MIMIC-IV in research.
 - **Redefining Model Success:**
 - **Fair performance, not just accuracy:** Ensures AI in healthcare supports ethical care for diverse populations.
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